

Git HandsOn - 3

Aim:

To understand Git branching and merging by creating a new branch, making changes, merging them into the main branch, and deleting the merged branch.

Objectives:

- Create a new Git branch.
- Switch between branches.
- Add and commit changes in a branch.
- Compare branches using git diff.
- Merge the branch into the main branch.
- View commit history using git log.
- Delete the merged branch.

Commands Used:

```
git branch GitNewBranch
git checkout GitNewBranch
git add branch.txt
git commit -m "Added branch.txt"
git checkout main
git diff main GitNewBranch
git merge GitNewBranch
git log --oneline --graph --decorate
git branch -d GitNewBranch
git push origin main
```

Outputs:

```
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ pwd
/c/Users/HP/Desktop/GitDemo

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ touch sample.log

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ mkdir logs

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ echo Error > logs/error.log

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ touch .gitignore

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ notepad++ .gitignore
bash: notepad++: command not found

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ code .gitignore

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore

nothing added to commit but untracked files present (use "git add" to track)

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git add .gitignore

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git commit -m "Added gitignore file"
[master 2bf59b3] Added gitignore file
 1 file changed, 3 insertions(+)
 create mode 100644 .gitignore

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
```

 MINGW64:/c/Users/HP/Desktop/GitDemo

```
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 305 bytes | 33.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/GayathriVanga/GitDemo.git
   5f4fe8f..2bf59b3  master -> master
```

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)

```
$ git status
On branch master
Your branch is up to date with 'origin/master'.
```

```
nothing to commit, working tree clean
```

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)

```
$ git status
On branch master
Your branch is up to date with 'origin/master'.
```

```
nothing to commit, working tree clean
```

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)

```
$ git branch
* master
```

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)

```
$ git branch GitNewBranch
```

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)

```
$ git branch
  GitNewBranch
* master
```

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)

```
$ git branch -a
  GitNewBranch
* master
remotes/origin/master
```

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)

```
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'
```

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)

```
$ git branch
* GitNewBranch
  master
```

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)

```
$ echo This is my new branch > branch.txt
```

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)

```
$ ls
branch.txt  logs/  sample.log  welcome.txt
```

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)

```
$ git add branch.txt
```

```
MINGW64:/c/Users/HP/Desktop/GitDemo
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ git add branch.txt
warning: in the working copy of 'branch.txt', LF will be replaced by CRLF the next time Git touches it

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ git commit -m "Added branch.txt in GitNewBranch"
[GitNewBranch e3c17da] Added branch.txt in GitNewBranch
1 file changed, 1 insertion(+)
create mode 100644 branch.txt

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git diff master GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..388da8a
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1 @@
+This is my new branch

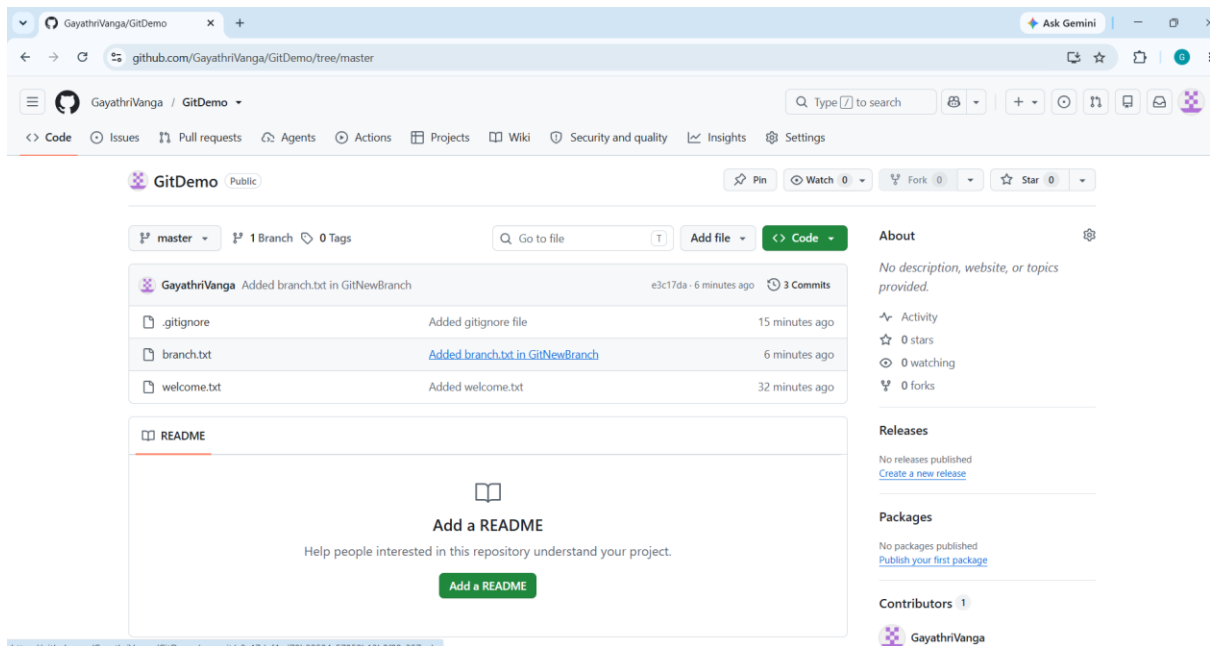
HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git merge GitNewBranch
Updating 2bf59b3..e3c17da
Fast-forward
 branch.txt | 1 +
1 file changed, 1 insertion(+)
create mode 100644 branch.txt

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git log --online --graph --decorate
fatal: unrecognized argument: --online

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was e3c17da).

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git branch
* master

HP@LAPTOP-JQJ9603M MINGW64 ~/Desktop/GitDemo (master)
$ git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 355 bytes | 71.00 KiB/s, done.
```



Result:

Successfully created a new branch (GitNewBranch), added a new file, committed the changes, merged the branch into the main branch, viewed the commit history, deleted the merged branch, and pushed the updated repository to GitHub.